

## FALCON's Response to Aramex for Supply of NEO – ASRS (1680 Bins)

February 16, 2024



Kind Attention –

Mr. Raad, Mr Yazan, Mr Zaid, Mr. Rami

M/s Aramex

**Offer Ref: F23-00313\_08 NEO-ASRS (1680 Bins); Date: 25 January 2024**

**Subject – Commercial Offer NEO - ASRS**

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

**Best Regards,**

**Sandeep Bansal**

**Chief Business Officer**

Contact Persons : Umer Saleem (Regional Head – ME)

Mobile : +971 (0) 507198540

Email : [umer.saleem@falconautoonline.com](mailto:umer.saleem@falconautoonline.com)

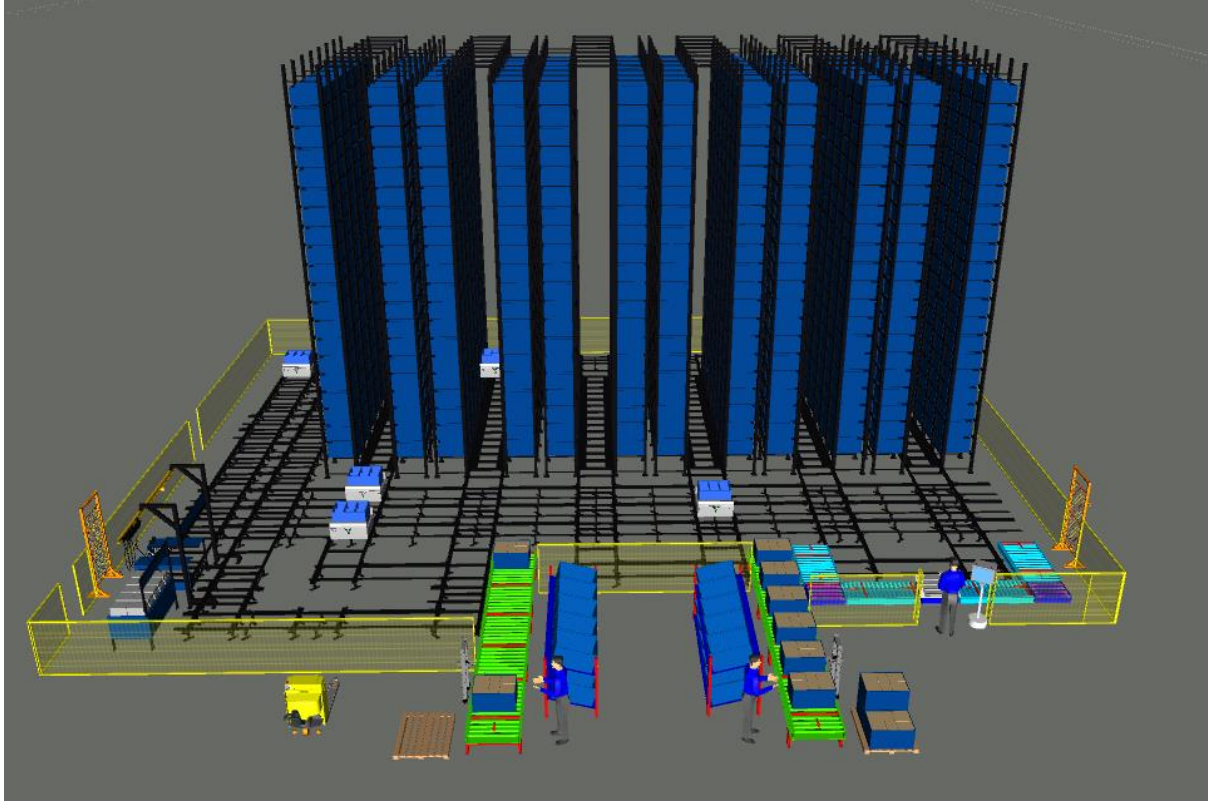
Company Name : Falcon Autotech Pvt. Ltd.

Regd. Office : 57, Nimri Colony, Double Storey Flats, Delhi – 110052

Website : [www.falconautotech.com](http://www.falconautotech.com)

LinkedIn ID : <https://www.linkedin.com/company/falconautotech/mycompany/>

## Response to RFP for NEO-ASRS\_ Aramex, Riyadh



Proposal Reference – F23-00313\_08

Falcon Autotech Private Limited  
Plot No. 87, Sector Ecotech-1, Extension-1,  
Greater Noida, Uttar Pradesh 201308.

Contact –  
Umer Saleem (Regional Head – ME)  
+971 (0) 507198540  
[umer.saleem@falconautoonline.com](mailto:umer.saleem@falconautoonline.com)

# Contents

|                                                                                    |    |
|------------------------------------------------------------------------------------|----|
| 1. Glossary .....                                                                  | 6  |
| 2. Executive Summary .....                                                         | 7  |
| 3. Company Profile .....                                                           | 8  |
| 4. Falcon's Experience and Achievements in Sortation Space Globally .....          | 12 |
| 5. Reference Projects .....                                                        | 14 |
| 5.1 Project 1- (CEP Client, UK) .....                                              | 14 |
| 5.2 Project 2- (CEP Client, Riyadh) .....                                          | 15 |
| 5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah) .....             | 16 |
| 5.4 Project 4- (CEP Client, Sydney) .....                                          | 16 |
| 5.5 Project 5- (Client – E-Commerce, India) .....                                  | 17 |
| 5.6 Project 6- (Ecommerce Client, India) .....                                     | 18 |
| 6. Handled Parcel Spectrum .....                                                   | 19 |
| 6.1 Parcel size loadable on the sorter .....                                       | 19 |
| 6.2 Parcels to be loaded on Sorter shall have the following characteristics: ..... | 19 |
| 6.3 Parcel not loadable on the sorter .....                                        | 20 |
| 6.4 Pictures of Sortable Parcels .....                                             | 21 |
| 7. Proposed System Description .....                                               | 22 |
| 7.1 Objective .....                                                                | 22 |
| 7.2 Summary of the System .....                                                    | 22 |
| 7.3 Main benefits of the NEO ASRS System .....                                     | 23 |
| 8. Concept Description .....                                                       | 24 |
| 9. Layout View of NEO ASRS .....                                                   | 25 |
| 10. Proposed System Capacity Calculations .....                                    | 26 |
| 11. System description .....                                                       | 26 |
| 11.1 NEObot .....                                                                  | 26 |
| 11.2 NEObin .....                                                                  | 27 |
| 11.3 NEOgrid .....                                                                 | 27 |
| 11.4 Conveyor based Station (Put) .....                                            | 28 |
| 11.5 Conveyor based Station (Pick) .....                                           | 28 |
| 11.6 System Components .....                                                       | 29 |
| 11.6.1 Accumulation Conveyor .....                                                 | 29 |
| 11.6.2 Powered Roller Conveyor .....                                               | 30 |
| 11.6.3 Side Barcode Scanners .....                                                 | 31 |
| 11.7 GTP Station .....                                                             | 32 |

|       |                                                                          |    |
|-------|--------------------------------------------------------------------------|----|
| 11.8  | Battery Charging Station .....                                           | 32 |
| 11.9  | NEOiT .....                                                              | 33 |
| 11.10 | Put to Light Module.....                                                 | 34 |
| 11.11 | Hand Held Scanner .....                                                  | 34 |
| 12.   | BOT Average Cycle Time .....                                             | 35 |
| 13.   | Conveyor BOQ .....                                                       | 36 |
| 14.   | Proposed System Technical Details of NEO ASRS .....                      | 36 |
| 14.1  | Mechanical equipment (NEO ASRS) .....                                    | 36 |
| 14.2  | Electrical, Controls & IT .....                                          | 36 |
| 15.   | Key Components Make .....                                                | 37 |
| 16.   | Program Organisation .....                                               | 37 |
| 16.1  | Program/ Project Schedule .....                                          | 37 |
| 16.2  | Program Management .....                                                 | 37 |
| 17.   | Aramex Responsibility.....                                               | 38 |
| 17.1  | Aramex Responsibilities During the Assembly and Commissioning Phase..... | 38 |
| 17.2  | Responsibilities of Aramex During the Tests.....                         | 38 |
| 17.3  | Aramex Responsibilities During the Training .....                        | 39 |
| 18.   | Commercial Terms.....                                                    | 39 |
| 18.1  | Price Sheet for NEO ASRS.....                                            | 39 |
| 18.2  | Payment Terms.....                                                       | 40 |
| 19.   | Warranty Period .....                                                    | 40 |
| 20.   | Exclusions .....                                                         | 41 |

## 1. Glossary

| S. No. | Term | Description                            |
|--------|------|----------------------------------------|
| 1      | RFP  | Request For Proposal                   |
| 2      | PPH  | Parcels Per Hour                       |
| 3      | ASRS | Automated Storage and Retrieval System |
| 4      | AC   | Alternating Current                    |
| 5      | DC   | Direct Current                         |
| 6      | PLC  | Programmable Logic Controller          |
| 7      | IT   | Information Technology                 |
| 8      | BOQ  | Bill Of Quantity                       |
| 9      | I/O  | Input/ Output                          |
| 10     | PDP  | Power Distribution Panel               |
| 11     | PC   | Personal Computer                      |
| 12     | UPS  | Uninterrupted Power Supply             |
| 13     | MDR  | Motor Driven Roller                    |

## 2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this Aramex RFQ ASRS. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.

Following the same objective for the Riyadh system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **The NEO – ASRS is running with 5 NEObots & 1680 bin locations.**
- **Total 3 Stations – 1 Nos GTP station & 2 Nos. Conveyor-based stations.**
- **PTL System for segregating inbound volume (Optional).**

**A tailor-made and simple layout designed explicitly for Aramex.** The proposed layout is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our meeting.

### 1. **Falcon's reliable Parcel sortation systems**

These systems are globally being used by most innovative brands such as Amazon, Asendia, Fastway, Delhivery, Flipkart and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

### 2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the Aramex Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 24 months from installation go-live.

### 3. **A strong and already successful partnership between Aramex & Falcon**

Aramex has been Falcon's customer since long and using multiple Falcon Systems and now a new parcel sortation system of 16,000 PPH is being installed at the Sydney location of the Aramex global network. For each of these projects, several advanced tools to improve the service's quality and efficiency are provided: Service Desk, Hotline support, remote access, regular governance meetings and monthly reports.



### 3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

## Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1Million+ Sorts Per Hour

600+ Employees

#### Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

نون noon

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

G

SWIGGY

TITAN

Hindustan Unilever Limited

#### News & Articles

**FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING**  
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

**FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP**  
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions  
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.



## Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

## Our Journey

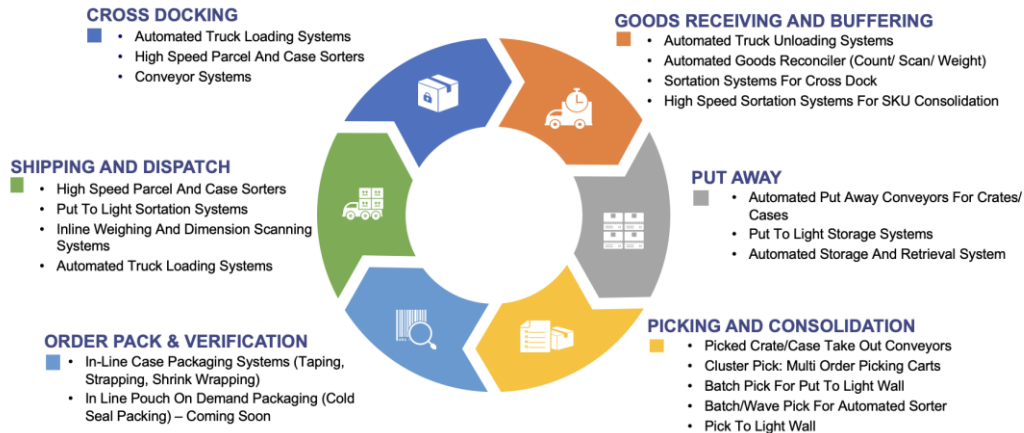
*Redefining Intralogistics Automation since 2012*



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

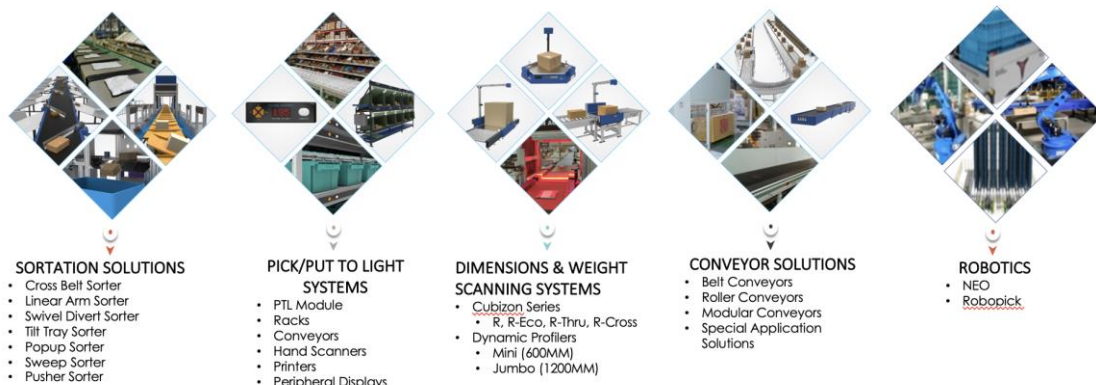
## Industry Process Solutions

*Delivering a multitude of game changing process automation solutions to our customers*



## Product and Solutions

*With 100% focused on Parcels, Eaches, Totes, Bags, Cartons*



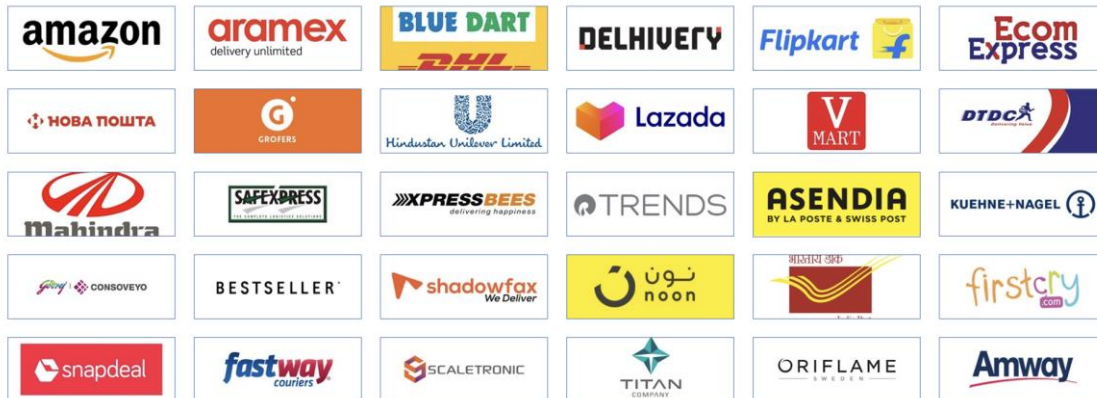
Powered by

Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

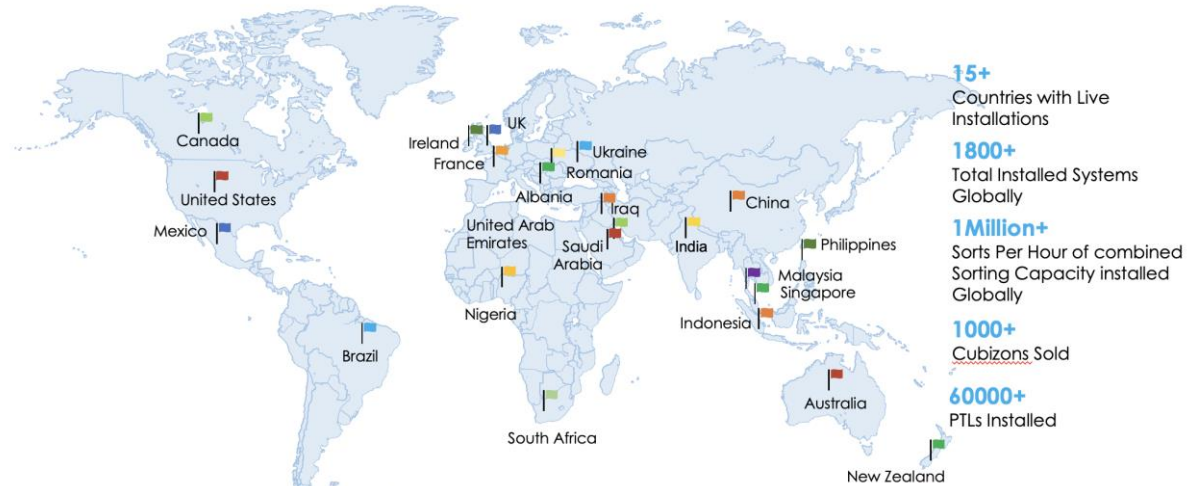
## Select Key Clients

*Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements*



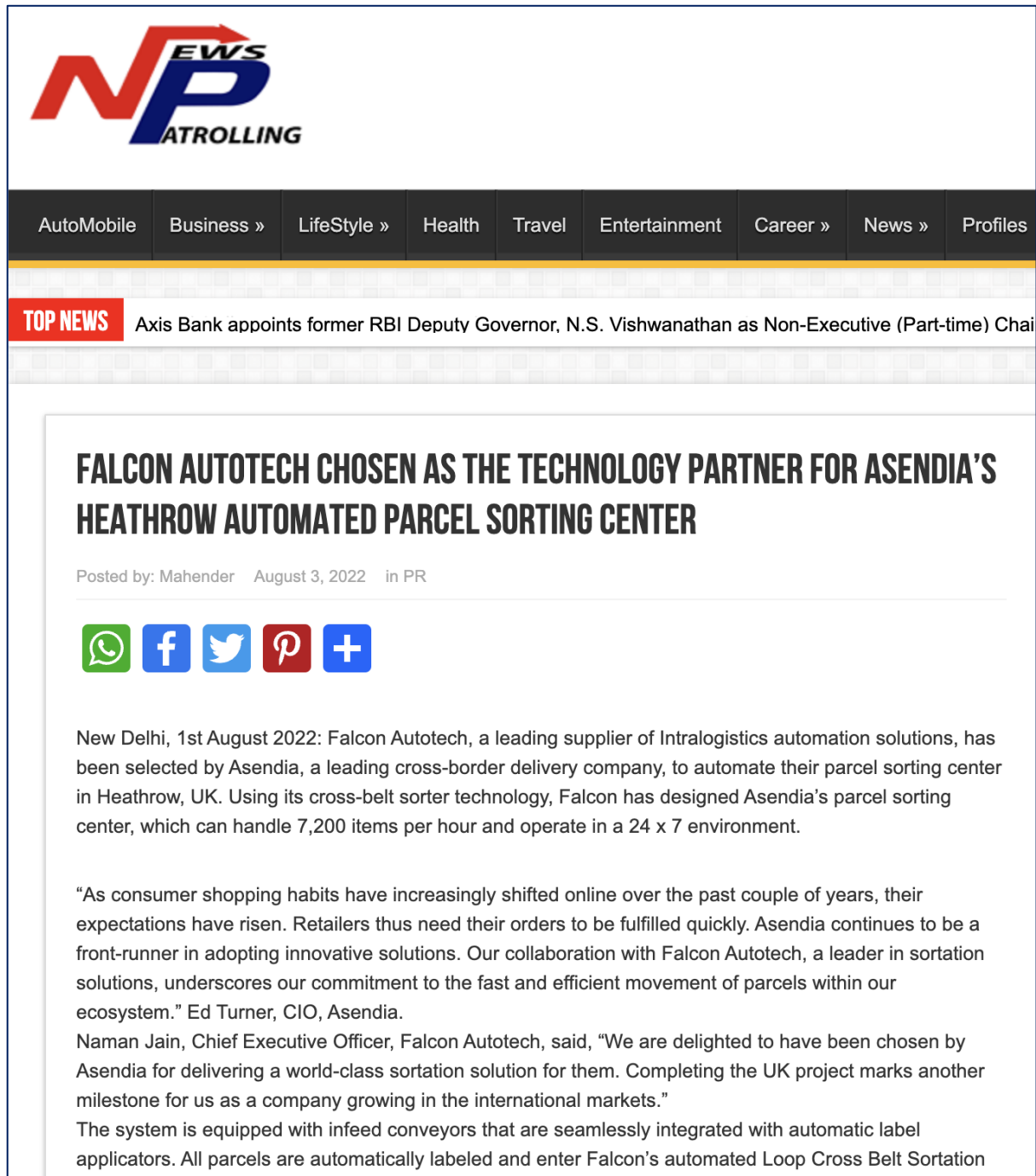
With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

## Global Market Presence



#### 4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



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## FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

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New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation



POSTED ON APRIL 17, 2023 IN / NEWS

## Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17<sup>th</sup> April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

## Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

## 5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

### 5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

#### Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

#### Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

#### Site Picture –





## 5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

### Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

### Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

### Site Picture –



### 5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

#### Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

#### Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

#### Site Picture –



### 5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

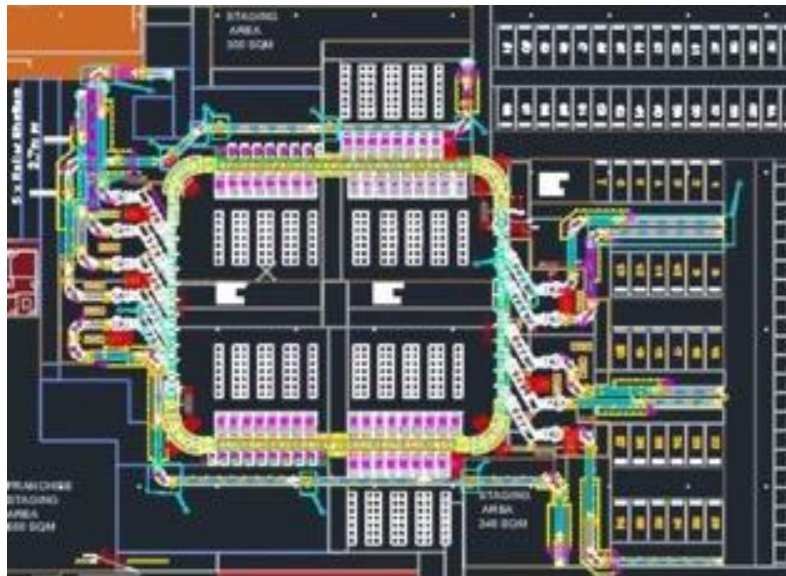
#### Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

#### Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

#### Site Picture –



### 5.5 Project 5- (Client – E-Commerce, India)

#### Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

#### Solution Specifications –

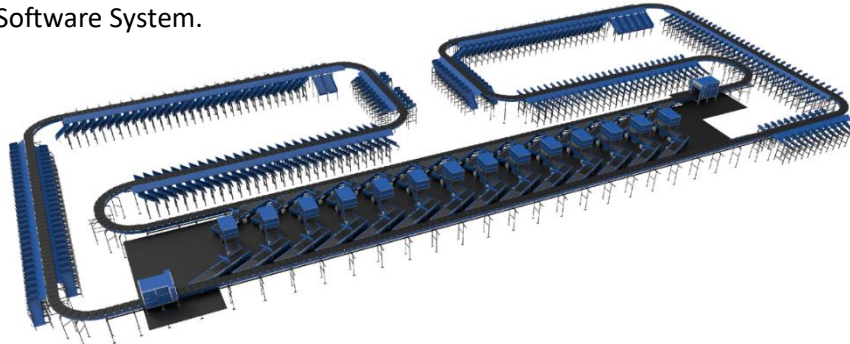
- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

#### Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.



- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



## 5.6 Project 6- (Ecommerce Client, India)

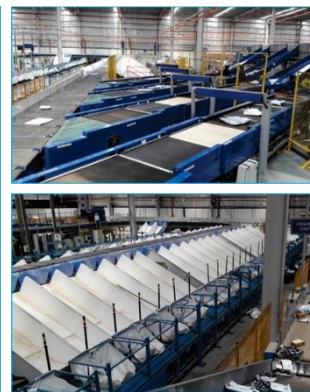
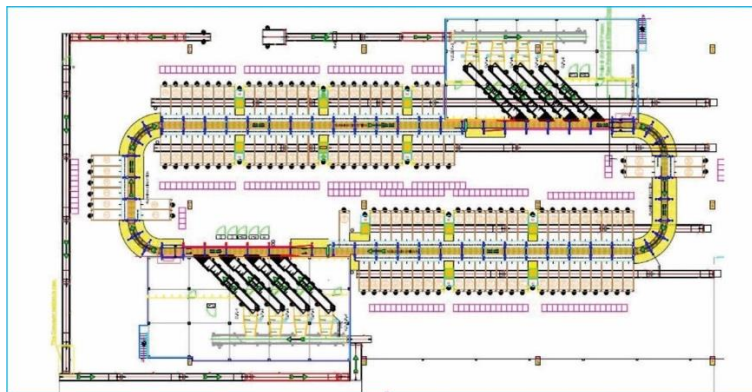
Use Case - Destination Sorting of Packed Shipments

### Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

**Key Technology Modules** - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

### Layout and Site Pictures -



## 6. Handled Parcel Spectrum

### 6.1 Parcel size

| Specification | Unit | Value |
|---------------|------|-------|
| Max Length    | mm   | 800   |
| Max Width     | mm   | 550   |
| Max Height    | mm   | 400   |
| Max Weight    | Kg   | 30    |
| Min length    | mm   | 100   |
| Min Width     | mm   | 100   |
| Min Height    | mm   | 3     |
| Min Weight    | gm   | 100   |

### 6.2 Parcels shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

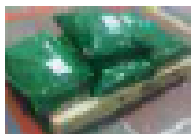
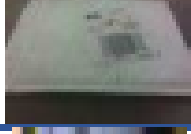


### 6.3 Parcel not loadable.

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

#### 6.4 Pictures of Sortable Parcels

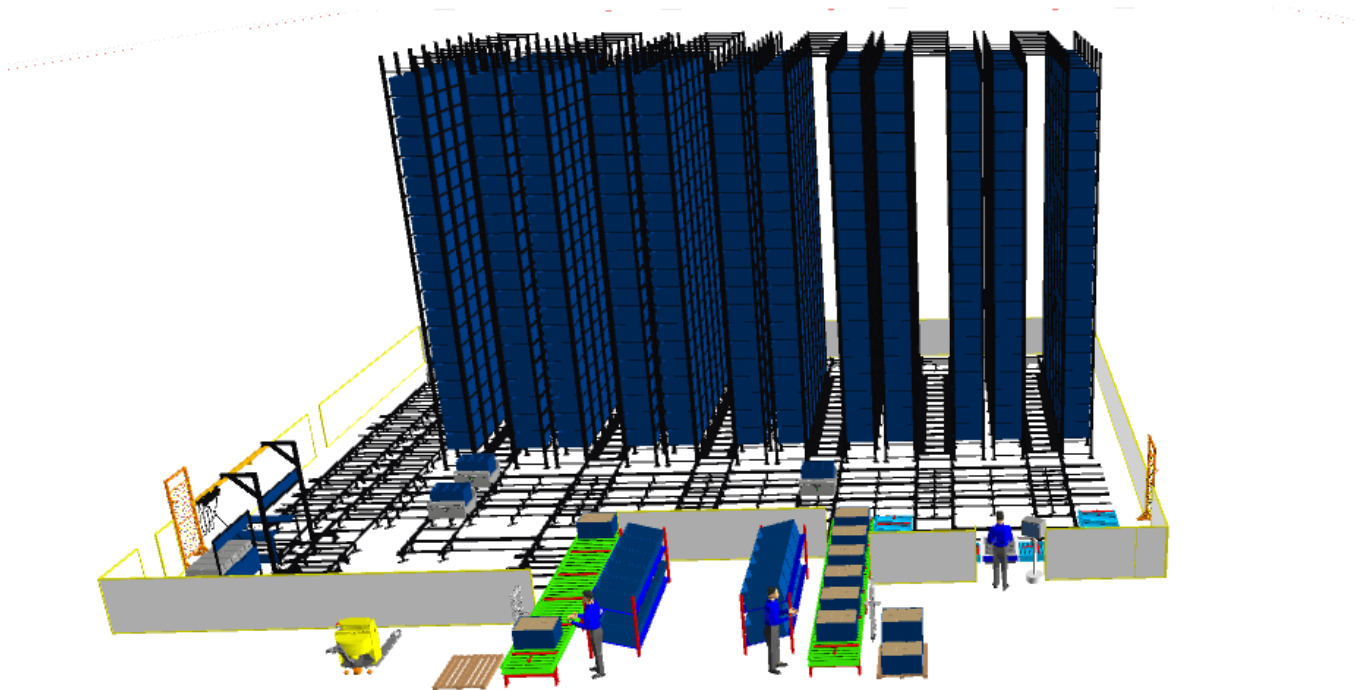
| PARCEL TYPE                                                                         |                                                    | DESCRIPTION                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Strongly Strapped                                  | A parcel banded with strapping material.                                                                                                                                                                              |
|    | Pre Manufactured Plastic Courier Pack <A3 in size  | A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.                                                                                                                        |
|    | Pre Manufactured Plastic Courier Pack > A3 in size | A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed. |
|    | Other Plastic Covered Item                         | A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.                                                                                                                 |
|   | Paper Covered Item                                 | A cardboard parcel coated in a packing paper.                                                                                                                                                                         |
|  | Cloth Covered Item                                 | Often Import parcels that are of irregular shape.                                                                                                                                                                     |
|  | Wine Box                                           | A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.                                                                                        |
|  | Cardboard Box                                      | A cardboard parcel is often the norm.                                                                                                                                                                                 |
|  | Totes                                              | A large plastic, lidded box.                                                                                                                                                                                          |
|  | Flats                                              | A parcel that often contains documents.                                                                                                                                                                               |
|  | Poly Bags                                          | It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.                                                                                                      |

## 7. Proposed System Description

### 7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of NEO - ASRS system to handle Return parcels.

### 7.2 Summary of the System



Process flow of the NEO ASRS System-

- a. **Infeed System:** - Parcels sorted from PTL system (Optional) SLA wise (Return in 7 days, 15 days, 21 days etc.) are manually placed inside a NEObin. The Parcels are scanned and mapped with barcode present on NEObin. One NEObin will have return parcels for the same SLAs. The Bins are palettized and transported to Put away station with the help of forklift/ BOPT.
- b. **Put Away System:** - Bins are manually placed on the conveyor of Put-away station. Bins are automatically scanned via side barcode scanner & travel further to the end of the conveyor, where a preassigned NEObot will pick the respective bin stored it at a defined bin location.
- c. **Retrieval System (Conveyor Station & GTP Station):** - Based on the SLAs of the return parcels on the respective day, the parcels will be retrieved from the ASRS. The Bots will retrieve the Bins containing the parcels from the storage locations and drop them on Conveyor Station for picking. An additional GTP station is also provided for Adhoc retrieval of specific parcels, requested by customer for re-attempting the delivery. Bot will bring the Bin containing that parcel and image of

the parcel will be displayed on the screen. Operator will pick and scan the parcel and transport it to CBS area for Sorting.

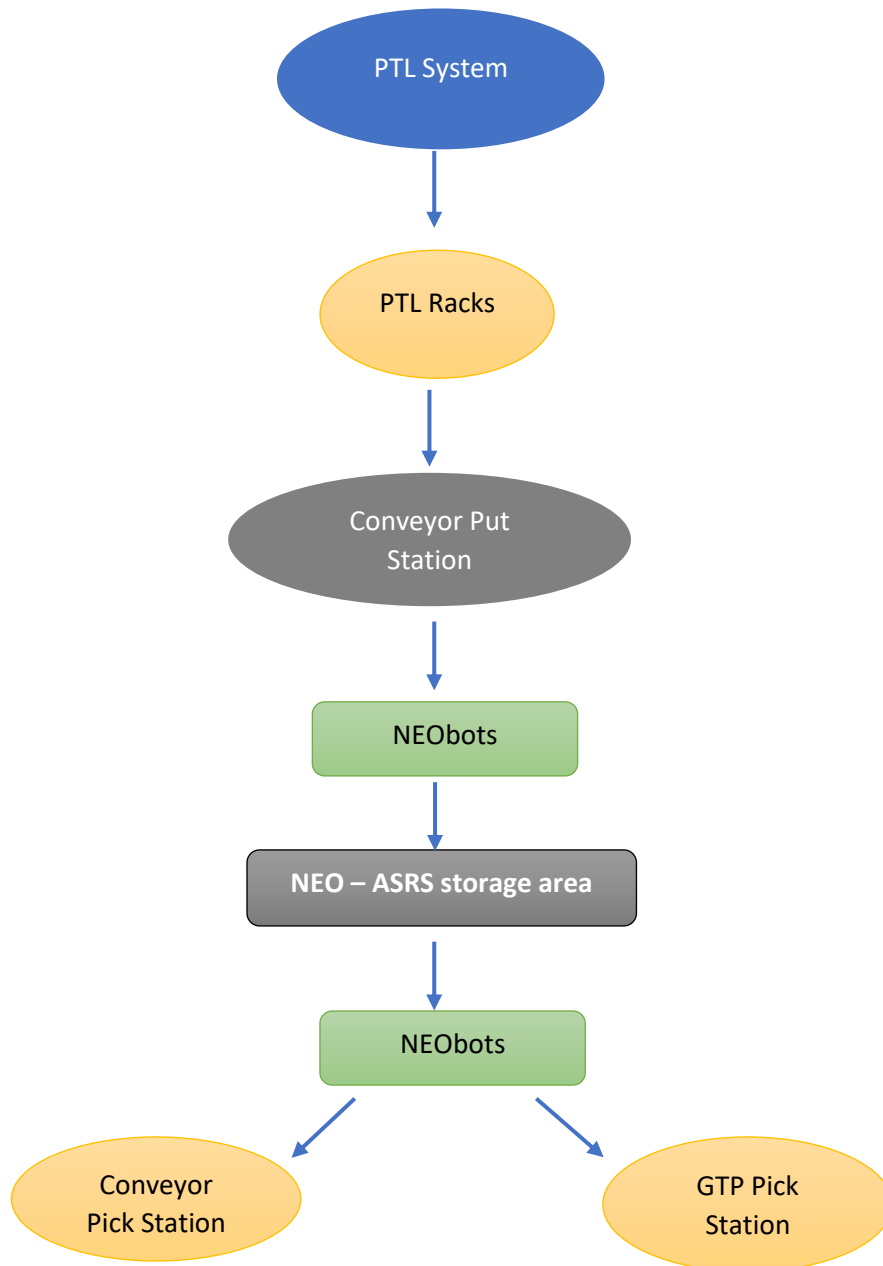
- d. **Outbound System:** - NEObins containing return parcels picked from the end of Conveyor station are palletized, transported and staged near CBS Infeed Area. These parcels are sorted shipper wise on CBS at the allocated time.

### 7.3 Main benefits of the NEO ASRS System

Falcon's Automated Storage and Retrieval System (ASRS) NEO offers a multitude of benefits over traditional storage systems, revolutionizing the way goods are stored and retrieved within warehouses.

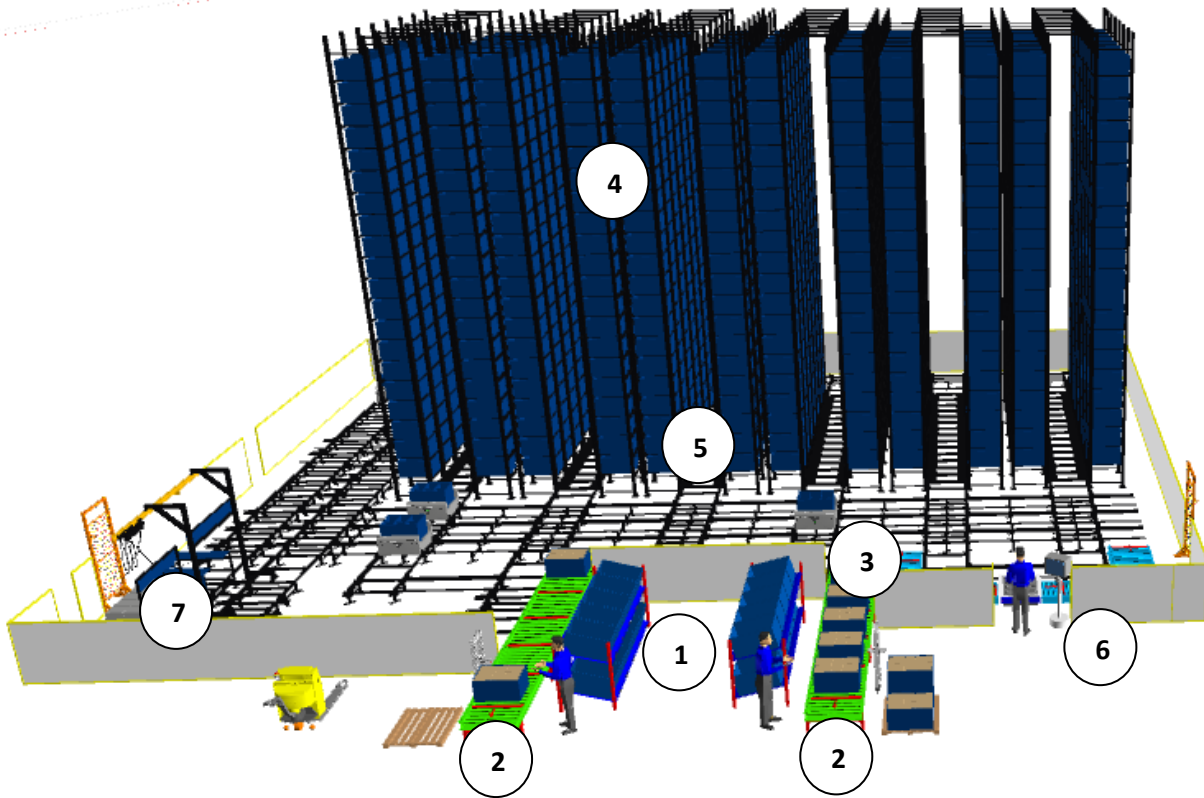
- **Enhanced Productivity:** NEO enhances your operator productivity by 4-5X. By eliminating the wait time, walk time and search time from picking operations.
- **Space Optimization:** NEO storage grid can go up to 16 meters to maximize vertical space utilization in your warehouse, allowing you to store more products in a smaller footprint and significantly increasing your warehouse's storage capacity.
- **Fast and Accurate Retrieval:** With automated picking and retrieval, NEO ensures lightning-fast and error-free order processing. This means you can meet tight delivery schedules with confidence and exceed your clients' expectations.
- **Inventory Management:** Gain real-time visibility into your inventory levels, helping you make informed decisions.
- **Scalability:** Whether your warehouse is small, large, or somewhere in between, NEO is scalable to adapt to your evolving business needs.

## 8. Concept Description





## 9. Layout View of NEO ASRS



### Legend

- 1 – PTL System
- 2 – Conveyor based Stations
- 3 – NEObot
- 4 – NEObins
- 5 – NEOgrid
- 6 – GTP station
- 7 – Charging station

## 10. Proposed System Capacity Calculations

The following table shows the calculation for the storage & retrieval system designed based on Aramex RFP requirements.

| PARTICULAR                                       | VALUE |
|--------------------------------------------------|-------|
| Total Storage (Bins)                             | 1680  |
| Average Number of parcels per Bin                | 12    |
| Number of Bots Required                          | 5     |
| Number of Conveyor based stations for Pick & Put | 2     |
| Total GTP Stations                               | 1     |

**Note- Average size of parcels considered – 260x210x170 mm.**

**Average number of bins 12, 70% utilisation, as agreed with Aramex over technical workshop call 11 Oct 2023.**

## 11. System description

### 11.1 NEObot

NEObot is your ultimate warehouse companion that never sleeps or is on leave. A powerful machine with minimum moving parts, NEObot is always ready to deliver at its best. Exquisitely designed and engineered, the bot requires no monitoring while at work. Using advanced Lidars and AI algorithms, it moves independently in the storage grid without continuous control from the software.e.

#### Technical Specification

**Weight:** 200Kgs

**Payload Capacity:** 40Kgs

**Speed:** 2m/s

**Size:** 1016 x 908 x 469 mm

**Operating temperature:** 0-40 degrees C

**Juice:** Swappable LiFepo4 battery



### 11.2 NEObin

Products are stored and moved in specialized bins or totes designed to be handled by NEO system. Each bin can be partitioned into 2/3/4/6/8 sub-sections for better product segregation.

#### Technical Specification

**Partitions:** 2/3/4/6 sub-sections

**Bin volume:** 5.75 cubic feet

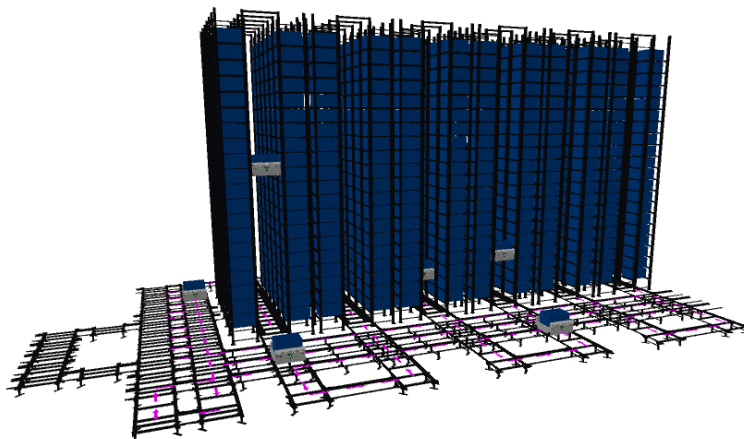
**External Dimension:** 810\*568\*425 mm



### 11.3 NEOgrid

NEOgrid is the steel structure which stores all your goods. With carefully crafted racks and tracks, the grid provides easy access to bots for storing and retrieving bins, cartons or trays. Based on the urban city layouts, NEOgrid combines interior tracks, highway tracks and superhighway tracks inside aisles for bots' movement between blocks to the grid and workstations.

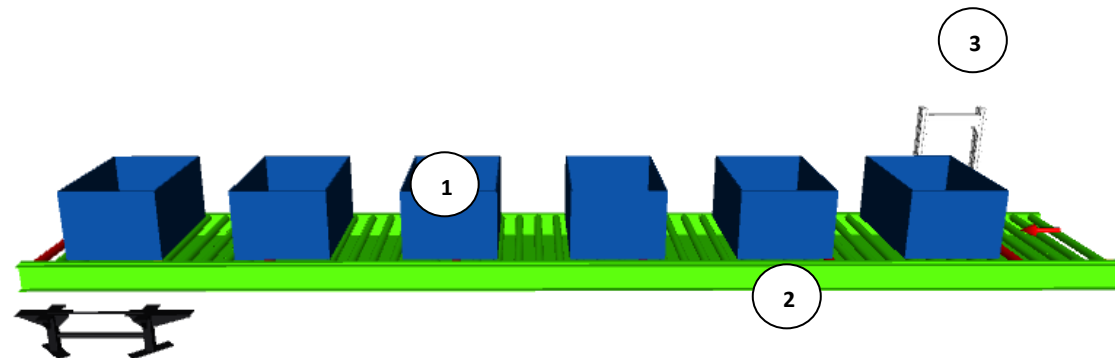
When a NEObot picks up a NEObin (which is waiting at the pick station), the system automatically assigns an appropriate rack inside the system, and NEObots proceed to that rack position to store the chosen bin.



### 11.4 Conveyor based Station (Put)

NEObins, (Containing return parcels sorted from the CBS) are transported from CBS to the NEO area using a combination of forklifts and manual handling. These NEObins are then placed onto a Conveyor Station. Subsequently, the NEObins are scanned using a side barcode scanner and proceed forward to be picked by one of the NEObots within the system for further processing.

2 Nos Conveyors based stations are used for Pick and Put operations.



#### Legend

- 1 – NEObins
- 2 – Accumulation conveyor
- 3 – Side barcode scanner

### 11.5 Conveyor based Station (Pick)

Once the parcels as per the SLAs are required to be retrieved on a particular day, the Bins containing these parcels will be called from Storage for outbound. The Bots assigned to retrieve these Bins, will pick the bins from storage locations and drop them on Conveyor Station.

NEO Conveyor stations are designed to keep the process simple. The NEObot drops the bins on the conveyor, which takes the bins to the operator working outside the grid. The bins are buffered on the conveyor to ensure wait-free picking operations, opening the horizon for multiple use cases.

Bins flow forward and gather at the end, ready to be picked up by an operator and palletized. Palletized NEObins are returned to the CBS infeed area, where they are manually put into the infeed conveyor at the allocated time for Shipper wise sorting.

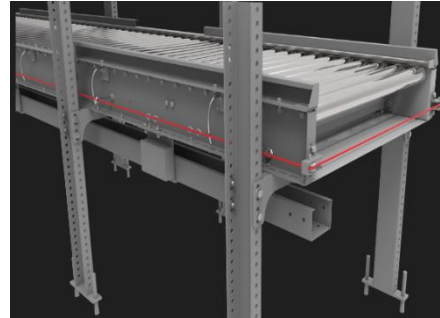
## 11.6 System Components

### 11.6.1 Accumulation Conveyor

Falcon's Accumulation conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's accumulation conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



| Specification                             | UOM  | Remark                                                                                               |
|-------------------------------------------|------|------------------------------------------------------------------------------------------------------|
| Manufacturer Name                         | Name | Falcon Autotech                                                                                      |
| Material of Roller                        | Type | MS Rollers with Zinc Plating                                                                         |
| Overall Width                             | Mm   | 900                                                                                                  |
| Conveyor Height                           | Mm   | As per Layout requirement                                                                            |
| Drive Power Rating                        | Watt | 40                                                                                                   |
| Type of Motor                             | Type | DC Rollers (MDR)                                                                                     |
| Ingress Protection                        | Type | IP 54                                                                                                |
| Type of Drive System                      | Type | Round/Poly V Belt Drive                                                                              |
| Insulation Class                          | Type | Yes                                                                                                  |
| Conveyor Speed                            | m/s  | Variable speed to meet TPH requirements                                                              |
| Load capacity per unit length of Conveyor | kg/m | 50                                                                                                   |
| Roller Span                               | Mm   | Approx. 600                                                                                          |
| Roller Diameter                           | Mm   | Approx. 50                                                                                           |
| Pitch of Rollers                          | Mm   | 75                                                                                                   |
| Zone Length                               | Mm   | 825                                                                                                  |
| Type of Mounts                            | Type | Fixed Heights Legs with Grouting Provisions                                                          |
| Type of Guides                            | Type | Steel guides at 50mm height basis the position of conveyor on both sides from Conveyor Belt Surface. |

### 11.6.2 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



| Specification                         | UOM  | Remark                                               |
|---------------------------------------|------|------------------------------------------------------|
| Manufacturer Name                     | Name | Falcon Autotech                                      |
| Material of Roller                    | Type | MS Rollers with Zinc Plating                         |
| Overall Width                         | mm   | 900                                                  |
| Conveyor Height                       | mm   | Variable as per design shared                        |
| Drive Power Rating                    | Watt | 40                                                   |
| Type of Motor                         | Type | DC Rollers (MDR)                                     |
| Ingress Protection                    | Type | IP 54                                                |
| Type of Drive System                  | Type | Poly V Belt Drive                                    |
| Insulation Class                      | Type | Yes                                                  |
| Conveyor Speed                        | m/s  | Variable speed to meet TPH requirements              |
| Load capacity/unit length of Conveyor | kg/m | 50                                                   |
| Roller Span                           | mm   | Variable                                             |
| Roller Diameter                       | mm   | 50                                                   |
| Pitch of Rollers                      | mm   | 75/100                                               |
| Type of Mounts                        | Type | Fixed Heights Legs with required Grouting Provisions |



### 11.6.3 Side Barcode Scanners

System is integrated with automated inline side barcode scanning system capable for scanning 1D barcodes with high accuracy.



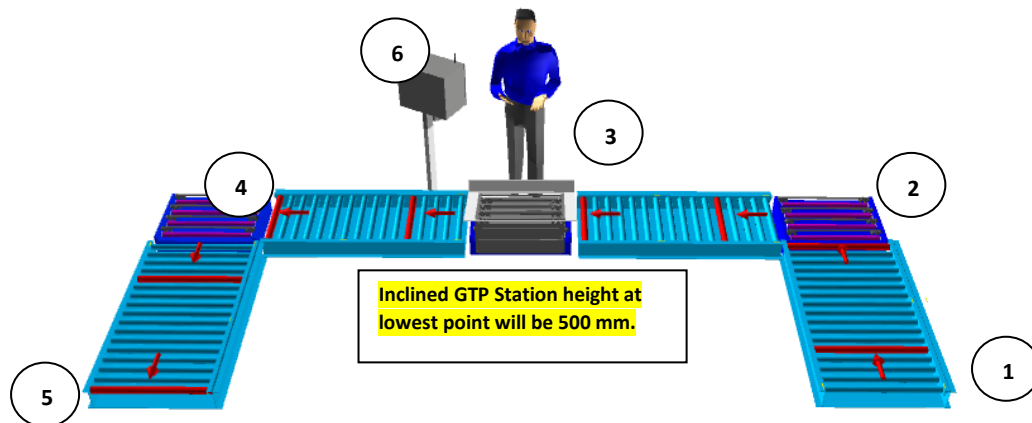
| Specification     | UOM  | Remark          |
|-------------------|------|-----------------|
| Manufacturer Name | Name | Cognex/Sick     |
| Barcode Type      | Type | 1D              |
| Module Size       | Mils | 12              |
| Number of Codes   | Nos  | 1               |
| Location          | Side | Left            |
| Orientation       | Type | Omnidirectional |

### 11.7 GTP Station

GTP station is also provided for Adhoc retrieval of specific parcels, requested by customer for re-attempting the delivery.

An operator may call the NEObin directly through the GTP station, or a predetermined logic may be configured in advance to retrieve a bin or collection of bins at a specific time interval on the day of breach of SLA.

Bot will bring the Bin having that particular parcel to the station and image of the parcel will be displayed on the screen. Operator will pick the parcel and transport it to CBS area for Sorting.

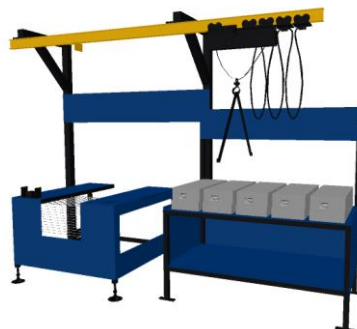


1. **Receiving module** – Bots will drop the bins at this position.
2. **Pop-up module** – This section will receive the bins and will transfer at 90°
3. **Tilter**- This is a tilter module which will receive the bins at 0° or flat condition then will tilt to  $\alpha^\circ$  and transfer to the next module.
4. **Exit pop-up module**- This section will receive the bins and will transfer at 90°
5. **Exit module**- This is the exit module from where the bot will pick up the bins.
6. **Screen**- Bot will bring the Bin having the particular parcel to the station and image of the parcel will be displayed on the screen.

*Note- Priority shipments, can be assigned to shipments requested by GTP over the standard inbound and outbound procedures.*

### 11.8 Battery Charging Station

It is the area where the battery of the shuttles is swapped and charged. NEObots with low volume hours can be replaced with new charged batteries or straight parked for charging.



*Note- Battery operational time is 6-8 hours (Fully charged), charge time is 2-2.5 hours (during non-operation hours)*

### 11.9 NEOiT

NEOiT is the brain of the NEO system. It is an in-house software stack to support all NEO operations, from order management to bot's route planning to bin allocations. Using robust AI algorithms and complex programming rules, NEOiT does all these operations like a breeze. Using API web services, it can quickly integrate with your WMSs, and with an intuitive user interface, it ensures faster and more efficient order preparations. NEOiT will ensure that the parcels storage inside the grid is based on the defined SLAs for shorter bot trips.



***Note- During non-operational hours, it is possible to move bins needed for the next day to lower levels within the ASRS racking.***

### 11.10 Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keep on altering the PTL configuration to increase the system productivity and give you the extra competitive edge.

The System involves a collection of physical sorting locations. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.



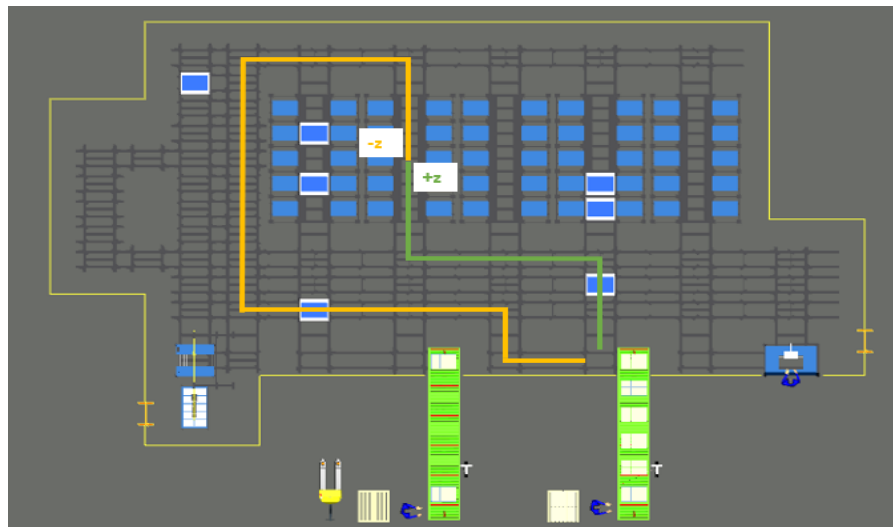
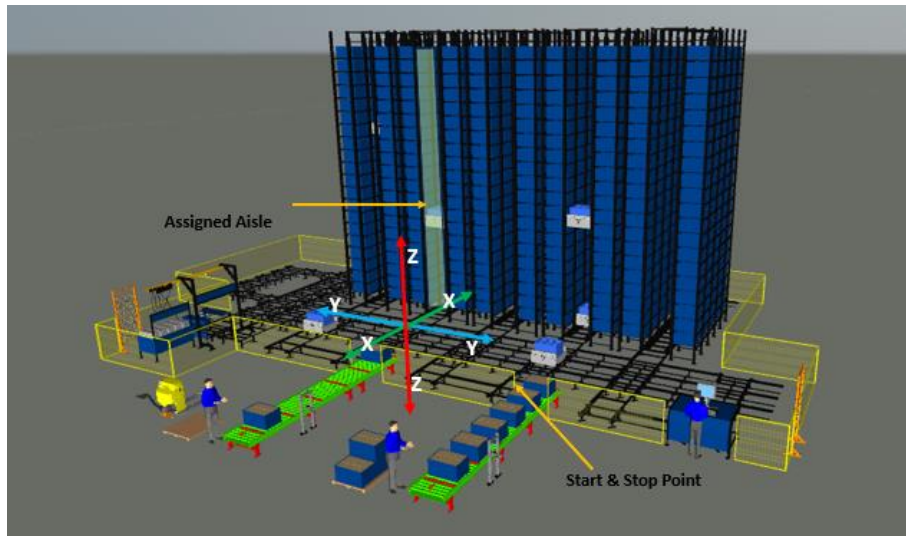
| Specification         | UOM    | Remark                                 |
|-----------------------|--------|----------------------------------------|
| Manufacturer Name     | Name   | Falcon Autotech                        |
| Dimensions            | Mm     | Around 165 X 50                        |
| PTL location          | Nos    | As per Layout requirement              |
| LED Display           | Type   | 3 Digit Alpha numeric Rotating Display |
| LED Size              | Mm     | ~ 25 X 20                              |
| LED Indicator         | yes/no | YES                                    |
| Acknowledgment Button | yes/no | YES                                    |
| Function keys         | yes/no | 4 Keys                                 |
| Power Requirements    | Type   | 3 Phase UPS Power                      |
| Communication         | Type   | TCP/IP and Serial                      |

### 11.11 Hand Held Scanner

Make – Motorola/ Honeywell/ Symbol



## 12.BOT Average Cycle Time



| S.No.                        | TIME TAKEN BY BOT IN A COMPLETE CYCLE OF PUTTING A BIS TO ALLOCATED POSITION | TIME TAKEN BY BOTS (SEC) |
|------------------------------|------------------------------------------------------------------------------|--------------------------|
| 1                            | Loading time of Bin on BOT                                                   | 8                        |
| 2                            | BOT Travel time till Location Tower                                          | 42                       |
| 3                            | BOT Tower climbing                                                           | 13                       |
| 4                            | Time required for Bin put at the Storage Position                            | 8                        |
| 5                            | BOT descend from tower and Axis Change                                       | 13                       |
| 6                            | BOT Moving to Initial position (Conveyor based Pick Station)                 | 66                       |
| Average Bot Cycle Time (Sec) |                                                                              | 150                      |
| Average Bot Cycle Time (Min) |                                                                              | 2.5                      |

**Note- 2.5 Mins is the average cycle time of Bots. With 5 Bots, we commit 120 Bins to be delivered on Stations. If we are not able to meet this Throughput, we will provide extra BOT at our cost. This calculation assumes that all the BOTs will be assigned for Put Away tasks from the Inbound Conveyor.**

### 13.Conveyor BOQ

| S No. | Name                         | EL_1 | Length(m) | Width(mm) | Set |
|-------|------------------------------|------|-----------|-----------|-----|
| 1     | Accumulation Roller Conveyor | 500  | 6.5       | 900       | 1   |
| 2     | Accumulation Roller Conveyor | 500  | 6.5       | 900       | 1   |
| 3     | Accumulation Roller Conveyor | 500  | 3.4       | 800       | 2   |
| 4     | Accumulation Roller Conveyor | 500  | 3.2       | 600       | 2   |
| 5     | Pop-Up Sorter Unit           | NA   | NA        | NA        | 2   |
| 6     | Pop-Up Sorter Unit (Tilted)  | NA   | NA        | NA        | 1   |

### 14.Proposed System Technical Details of NEO ASRS

#### 14.1 Mechanical equipment (NEO ASRS)

| Pos. | Qty. | Description                                                                                                                                                                                                                                                                                   | Value                                                                                            |
|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1    | 1    | <b>NEO - ASRS System</b> <ul style="list-style-type: none"> <li>Bin Locations</li> <li>Conveyor based stations.</li> <li>GTP Station</li> <li>Neo Bots</li> <li>Fast Charge Batteries</li> <li>Accumulation conveyor</li> <li>Fencing</li> <li>Gates</li> <li>Side Barcode scanner</li> </ul> | 1680 Nos<br>2 Nos<br>1 Nos<br>5 Nos<br>3 Nos<br>6 sets of 19.6 meters<br>1 Set<br>2 Nos<br>2 Nos |
| 2    | 1    | <b>PTL System (Optional)</b> <ul style="list-style-type: none"> <li>PTL Rack bin type</li> <li>PTL Modules</li> <li>Hand barcode Scanner</li> </ul>                                                                                                                                           | 20 Nos.<br>20 Nos.<br>02 Nos.                                                                    |

#### 14.2 Electrical, Controls & IT

| Electricals |   |                                                                                                                                                                                                                                                                                          |          |
|-------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1           | 1 | <b>Field wiring, control &amp; IT</b> <ul style="list-style-type: none"> <li>Main Control Panel</li> <li>Network Switches</li> <li>Field Cabling</li> <li>Barcode Scanner Controls</li> <li>Conveyor Controls</li> <li>IT</li> <li>Control IT Software Package</li> <li>NEOiT</li> </ul> | Included |



## 15.Key Components Make

| Items          | Make                     |
|----------------|--------------------------|
| Rollers        | Falcon                   |
| Encoders       | SICK                     |
| Sensors        | Sick/Leuze/ P&F          |
| PLC            | Siemens                  |
| Control Panels | Rittal/ BCH              |
| VFDs           | Siemens/Lenze/ AB/ Omron |
| Cables         | LAPP/ Equivalent         |
| Switch Gear    | Schenider/Equivalent     |
| Bearings       | NTN/SKF/Equivalent       |
| HMI's          | Siemens                  |
| MDR            | Pulse/ Itoh Denki        |

## 16. Program Organisation

### 16.1 Program/ Project Schedule

| S.No. | Activity                    | Timeline |
|-------|-----------------------------|----------|
| 1.    | Receipt of PO               | Week 0   |
| 2.    | Project Kick off            | Week 1   |
| 3.    | DAP Submission              | Week 3   |
| 4.    | DAP Approval                | Week 5   |
| 5.    | Detailed Designing          | Week 8   |
| 6.    | Manufacturing & Procurement | Week 14  |
| 7.    | Inhouse Assembly            | Week 18  |
| 8.    | Dispatch                    | Week 19  |

Transit time- 5 weeks

**Installation & Commissioning- 5 to 7 weeks.**

*\*Above plan is tentative, detailed plan will be shared post order confirmation.*

### 16.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Aramex
3. Weekly/Fortnightly Meeting Aramex to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

## 17. Aramex Responsibility

### 17.1 Aramex Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- Aramex to provide 1 nos. forklift for 60 days.
- HPT/BOPT & Scaffolding - 3 nos. each is considered in customer scope for throughout the project duration.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the parcelsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

### 17.2 Responsibilities of Aramex During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the parcels correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

### 17.3 Aramex Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough space for desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technical staff.
- The invitation of staff to attend the training courses will be at the expense of ARAMEX.

## 18. Commercial Terms

### 18.1 Price Sheet for NEO ASRS

| S. No       | Component                | Set | Price (USD) |
|-------------|--------------------------|-----|-------------|
| 1           | Rack + Bins              | 1   | \$ 100,734  |
| 2           | Bots                     | 1   | \$ 102,651  |
| 3           | Other Infra (Neo System) | 1   | \$ 81,685   |
| 4           | Conveyor Automation      | 1   | \$ 39,703   |
| 5           | Software Packages        | 1   | \$ 8,212    |
| 6           | Toolkit                  | 1   | \$ 1,092    |
| 7           | Packaging & forwarding   | 1   | \$ 16,518   |
| 8           | Project Management + I&C | 1   | \$ 117,077  |
| Total (USD) |                          |     | \$ 467,673  |

**Extended Warranty 1 Year included above price.**

Additional Price for PTL System \$ 7,205

Additional Price for a BOT \$ 22,500

Additional Price for a Fast Charge Battery \$ 1,950

Additional price for HHT based Sorting \$ 2,600

**Note- HHT (Handheld Terminal) based sorting considered, HHT will be in client's scope.  
App deployment and Integration charges included in Falcon scope.**

The Total Price is to be understood.

- Freight – Excluded
- Taxes as applicable
- Packing and Forwarding - Included
- Installation and Commissioning - Included
- Price is valid for 60 days from the date of proposal.

\*Material Unloading & Laydown area is in customer scope.

Spares not included above for client consideration:

**Critical Spares \$24,130 (Includes 3 Nos. Fast Charge Batteries)**

## 18.2 Payment Terms

| Payment Percentage | Stage                             |
|--------------------|-----------------------------------|
| 25%                | Advance along with PO             |
| 35%                | Against prior dispatch against PI |
| 15%                | Material received at Site         |
| 15%                | After Installation and Trial test |
| 10%                | Acceptance and complete handover  |

*\*All invoices (if due as per payment terms) should be cleared within 7 days from the date of receipt of undisputed invoice by the buyer*

## 19. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use) and post completion of warranty.

1-year extended warranty included.

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.

- Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
- A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

## 20. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Server System & SCADA
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system
- Any Kind of Civil work
- Fire-alarm and fire protection devices
- Electrical power supply and wiring to the main control cabinets
- Construction Power, temporary power during installation
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Traffic and route markings
- Laydown Area, with security / watchman, if required, any special gate passes at cost
- Assembly tools like forklifts and hoisting machines
- Steel Works- If not specified, ram protection devices, cat walks, bridges, maintenance aisles and platforms
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines (if applicable)
- Roller cages/ Pallets (if applicable)
- **Simulation and 3D animation.**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.